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A Novel Distribution-Driven Federated Framework for Evaluating Healthcare IoT Data Anonymization under Privacy–Utility Equilibrium

Abstract- The explosion of the Internet of Things (IoT) in healthcare has offered parallel opportunities as well, such as continuous patient monitoring and data-driven clinical decisions; however, it also raised concerns about privacy. There are many anonymization methods available for securing sensitive healthcare IoT data and their evaluations have been mostly performed discretely in prior work, with only a limited number of studies quantifying privacy–utility trade-off in an integrated and clear manner. This paper presents HAPED-QA-FL (Healthcare Anonymization Performance Evaluation using Distributional Quality Assessment with Federated Learning), an evaluation framework for privacy, where one does not leverage federated learning as a security tool but as a distributed analytics engine to evaluate the impact of anonymization on model utility across healthcare at scale without centralizing raw data. Anonymization is evaluated by the framework at two complementary levels, i.e., distribution-driven analysis that employs statistical divergence, entropy-based information loss, and quasi-identifier uniqueness; coupled with federated performance validation in which anonymized datasets are collaboratively analyzed to infer utility loss and re-identification risk under realistic IoT healthcare scenarios. A generalised privacy–utility equilibrium score (PUES) is proposed to rank anonymisation schemes with respect to their data-utility preservation, information loss and the re-identification risk. Experiments conducted on real-world healthcare IoT datasets show that the introduced framework provides a more stable and interpretable solution method to anonymization evaluation over traditional single-metric based solutions, providing a scalable and domain-adherent way for privacy support in healthcare IoT data management.

Keywords: Healthcare IoT, Data Privacy, Data Anonymization, Federated Learning, Privacy–Utility Trade-off, Information Loss, Re-identification Risk, Distributional Assessment

I Introduction

The introduction of the Internet of Things (IOT) in Healthcare changed how clinical data are acquired allowing for continuous monitoring, personalised treatment and real-time decision support [1][2]. Wearable sensors, intelligent medical devices and remote patient monitoring systems have created the flood of heterogeneous data streams that move across hospitals [3] diagnostic centers [4] and cloud platforms. Although these features enhance the quality of life for patients, they also lead to a new degree of exposure of personal health data [5][6]. Even when direct identifiers do

not exist in the released data, combinations of quasi-identifiers can still enable an individual to be re-identified, resulting in privacy threats being long-term and hard-to-treat [7]. Given that healthcare facilities are more and more relying heavily on big data analytics, there is an urgent need for evaluation frameworks that guarantee privacy while preserving the analytic value of data [8][9].

Several research works have suggested anonymization techniques to mitigate privacy risks in healthcare IoT data but they were evaluated separately and by using incomplete features [10][11]. Some assessments consider only privacy and do not consider how much utility is lost of the data, while others use centralized learning techniques that are impractical if agencies can't share their data. Therefore, the trade-off between data utility [12][13], information loss and reidentification risk is not easily quantified by one easy-to-understand metric [14][15]. To address these problems, we present HAPED-QA-FL, a distribution-supported framework for quantitatively assessing anonymization process. The framework also does not adopt federated learning as a security technique or for collaborative computation. This permits the evaluation of the impact of anonymization on model performance in a multi-institutional setting without sharing raw data across sites [16][17]. Through the integration of distribution-based metrics and federated validation, this approach offers a straightforward and operationally feasible path for selecting anonymization methods that ensure privacy without compromising valuable clinical information [18][19].

1.1 Background & Motivation

Healthcare IoT environments gather ongoing physiological, behavioural and contextual information which constitute extremely sensitive details of patient's life. The traditional anonymization research works hardly consider their distributed, high-dimensional [20][21] and dynamic aspects. We thus need a structured, reproducible methodology to assess anonymization methods under realistic health care deployment scenarios.

1.2 Privacy problems in Healthcare IoT

However, anonymized datasets are also likely to be privacy-threatening as individuals can still be re-identified through linkage attacks [21], quasi-identifier matches [22], and distributional inference. The risk is multiplied when data is shared between different entities and used together on analysis – resulting in privacy assurance being more of a continuous process rather than just a one-off exercise.

1.3 Limitations of existing anonymization evaluations

Most prior evaluations:

- Analyze anonymization techniques independently,

- Rely on single privacy metrics,
- Or require centralized datasets.
- Such perturbations do not necessarily preserve the true privacy–utility trade-off and are not easily applicable in a federated healthcare setting.

1.4 Role of Federated Learning in evaluation

In this work, federated learning is not applied as a cryptographic or security tool. Rather it allows a distributed evaluation environment to calculate the impact of anonymization on utility at many healthcare sites without sharing raw patient information.

1.5 Major Contributions

- A distribution-based privacy evaluation framework (HAPED-QA-FL) for the healthcare IoT.
- Privacy-utility equilibrium score (PUES) It is a Unified Privacy Utility Equilibrium Score considering utility, information loss and re-identification risk.
- Evaluation with two levels: Statistical distribution and analysis federated performance validation..
- Transparent, readable approach to comparing anonymization techniques in the real distributed healthcare setting.
- Data-driven explanations for how anonymization decisions impact downstream analytics.

1.6 Organization of the Paper

The rest of this paper is organized as follows. Section 2 introduces the main anonymization and federated learning methods for healthcare IoT data, focusing on their privacy–utility tradeoffs. In Section 3, we describe the details of our proposed HAPED-QA-FL and how we evaluate through evaluation methods to demonstrate anonymization utility. The data, evaluation measures and experimental setup we used for empirical evaluation are described in Section 4. Section 5 presents the comparative results, and examines how anonymization strategies perform in our evaluation model. Results And also discusses practical implications and presents future directions for research. Finally, Section 6 concludes the paper and discusses the contribution of the present study.

II Literature Review

2.1 Taxonomy of Privacy-Preserving Healthcare IoT Analytics

According to the surveyed works, privacy-preserving healthcare internet-of-things analytics is systematically categorized into four primary aspects:

- (i) classical data anonymization techniques,
- (ii) utility-aware anonymization enhancements,
- (iii) distributed evaluation using federated learning, and
- (iv) privacy–utility trade-off assessment mechanisms.

This classification represents the progress from static anonymization to distributed, performance-based systems suited for actual deployments of healthcare IoT.

2.2 Classical Anonymization Techniques for Healthcare Data

Early and research work focus on data anonymization as the mechanism to ensure privacy protection in healthcare datasets. Khivsara et al (2025) provide an extensive report on techniques for privacy preservation in healthcare, and specifically focus on attribute centric anonymization and synthetic data methods. They demonstrate that the techniques that generate data in terms of attributes are more effective in maintaining original data integrity and ensuring analytic value as opposed to those that only rely on synthetic-data resolution [1]. The authors suggest that an attribute-oriented approach to anonymization provides a more predictable balance of privacy and utility for the analysis of healthcare data such as those used in their experiments.

Extending the classical approach, Karagiannis et al. [2] concentrate on k-anymity as a well-established anonymization method in healthcare to allow sharing of medical data. Their study shows how organized generalization and suppression minimize re-identification risk but shows limitations in terms of data manipulation, and reduced utility when the hamper threshold for anonymization increases.

Ren et al. [3] incorporate anonymization into IoT data processing for analytics, classifying health-critical IoT streams as contextual, continuous and media. Their findings demonstrate the potential of anonymization for privacy coverage without excessive loss to utility, yet also show the performance of anonymization is highly dependent on the data modalities.

Together, these studies demonstrate that anonymization is an essential layer of privacy but reveal a key gap the lack of a common standard for evaluation across datasets, domains, and analytic goals.

2.3 Utility-Aware and Attribute-Focused Anonymization

A number of works acknowledge the shortcomings of classical anonymization and propose to improve on it with utility-awareness. Srijayanthi and Sethukarasi [15] present clustering-based anonymization and feature selection technique that proves

that elimination of redundancies attribute significantly reduces information loss preserving privacy. This work not only emphasizes the significance of attribute relevance in privacy-utility trade-off but also suggests that training over imbalanced private data should be approached with caution.

These methods represent a transition from privacy-only anonymization to more balanced anonymization, with analytic utility as a primary design consideration. However, such methods are mainly centralized and less applicable to the multi-institutional healthcare IoT systems.

2.4 Federated Learning as a Distributed Evaluation Paradigm

The framework of federated learning emerges as a critical enabler for distributed healthcare analytics that allows training models without accessing raw data. For instance, Liu et al.(2025) conduct a systematic review of deep learning-based synthetic data generation methods that employ GANs, VAEs, and diffusion models for the purpose of healthcare privacy preservation. The researchers show that properly-trained synthetic data may effectively reduce re-identification risk while still sharing statistical similarity to real data, yet challenges still exist for temporal or high-dimensional medical data. They stress the necessity of evaluation standards that would allow combining privacy and data utility analysis . Another two systematic reviews conducted by Li et al. suggest that FL indeed enables collaborative healthcare modeling without sharing the actual data. Desirably, both of these works mentioned that the model utility fully depends on the data preprocessing and make-no-harmfully changes. It is re-emphasized by Mohammadi et al. and Pati et al. who additionally highlight the inevitability of quantitative privacy-utility trade-offs in FL systems which could harm accuracy, convergence time, and model size. More application-specific papers, such as Health-FedNet [12] or FedDeepInsight [13], actually demonstrate that FL is easily used in real healthcare applications. Although the systems were primarily developed to provide secure learning, they implicitly confirm that the transformation and preprocessing of data strongly affect the performance of federated models, positioning FL as an important evaluator of anonymized datasets.. Haripriya et al. suggest an adaptive federated learning framework that distinguishes between FedAvg and FedSGD aggregation based on the heterogeneity of the data [16]. The researchers show that the adaptive approach helps to improve the convergence efficiency and algorithm execution time, and maintains the high diagnostic accuracy on real distributed medical datasets. This work confirms that adaptive federated aggregation may be applicable for large-scale multi-center healthcare settings; however, they only consider its utility for model optimization but do not assess the actual transformation of data distributions or re-identification risk. In

general, FL is presented as preserving privacy of learning systems, but not yet utilized as an instrument to systematically evaluate the quality of data anonymization.

2.5 Privacy–Utility Trade-off and Evaluation Gaps

Privacy assessment is a fragmented domain overviewed from the literature—studies focused only on anonymization strength or federated model. Only a few works collectively analyze data utility, information loss, and re-identification risk under a unique framework. While Rahman et al. [8] and Saha et al. [10] discuss larger privacy issues in FL-based healthcare systems, they fail to provide a dataset level anonymization evaluation concept. Specifically, there exists no distribution-driven, federated, and metric-unified framework that objectively evaluates anonymization techniques under realistic healthcare IoT constraints. This directly motivates the proposed HAPED-QA-FL framework.

2.6 Literature Review Table

Table 1. Comparative Analysis of Privacy-Preserving Healthcare IoT Studies (2023–2025)

Ref	Year	Journal / Publisher	Domain	Technique Focus	FL Used	Privacy Metrics	Utility Consideration	Key Limitation
[1]	2025	Springer	Health care	Attribute-centric anonymization, synthetic data review	No	Re-identification risk, attribute disclosure	Yes	No distributed or federated evaluation
[2]	2024	Springer	Health care	k-Anonymity	No	k-value	Limited	Utility loss
[3]	2024	Springer	IoT	Stream anonymization	No	Re-ID risk	Yes	Not healthcare-specific

[4]	2025	Springer	Health care	Deep learning–based synthetic data generation (GAN, VAE, Diffusion)	Indirect	Privacy leakage, re-identification risk	Yes	Limited handling of temporal and high-dimensional data
[6]	2024	Springer	Multi-domain	FL privacy metrics	Yes	Accuracy, loss	Yes	Model-centric
[7]	2025	Elsevier	Health care	FL review	Yes	Performance bias	Yes	Clinical applicability
[10]	2024	Springer	General	FL privacy survey	Yes	Leakage risk	No	Security-heavy
[12]	2025	Elsevier	Health care	FL framework	Yes	Privacy guarantees	Yes	Encryption-dependent
[13]	2025	Elsevier	Health care	FL preprocessing	Yes	Accuracy, efficiency	Yes	Not anonymization-centric
[15]	2023	Elsevier	Health care	Feature-based anonymization	No	Info loss	Yes	Centralized
[16]	2025	Springer	Health care	Adaptive federated	Yes	Data exposure risk in distributed	Yes	Does not analyze data anonymi

				learning aggrega tion (FedAv g/FedS GD)		ed learning		zation techniqu es
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Table 1 depicts a comparative chart of recent privacy-preserving healthcare IoT studies published between 2023 to 2025. The comparative points to the primary research region, anonymization or federated learning techniques evaluation metrics for privacy assessment, and data utility preservation effort. The table distinctly distinguishes the classical anonymization studies, federated learning distributed model analytics, and privacy–utility trade-off works. Moreover, the current approaches limitations are summarized to present a lack of unified, distribution-driven evaluation frameworks. Overall, this comparative aspect of the existing scenario motives the need for the proposed HAPED-QA-FL framework that systematically enables anonymization assessment with federated evaluation under realistic healthcare IoT conditions.

2.7 Key Gaps Identified

The critical review of the most recent papers on anonymization and federated learning in healthcare IoT data pinpointed a number of prominent gaps. First, although many anonymization methods have been proposed (e.g., k-anonymity, clustering-based anonymization and feature selection), these methods are not well evaluated together in a holistic framework that also considers data utility, information loss, and re-identification risk. Therefore, comparison between anonymization algorithm is still non-standardized and data specific. Second, the current anonymization research mostly focuses on centralized evaluation conditions, which are not realistic for healthcare IoT systems in which data are geographically distributed among several organizations and cannot be shared due to legal or ethical reasons. This narrows the real-world application of these methods to large, multi-site healthcare systems. Third, though federated learning has been well studied as a privacy-preserving learning paradigm, most existing works focus on secure model training and parameter protection, rather than exploiting federated learning to be an objective evaluation vehicle for assessing the impact of anonymization such that it can affect analytical performance downstream. Therefore, the trade-offs between anonymization strength and federated model utility are not well studied.

Fourth, existing transform-based privacy analysis methods often rely on single- or limited-objective metrics (e.g., k-anonymity level and/or model accuracy), which cannot fully represent the balance point between all aspects of privacy–utility

tradeoffs. However, the lack of an embedded scoring function leaves it open to selection bias regarding de-identification strategies that are appropriate for healthcare IoT datasets. There is no distribution-driven evaluation of how anonymization changes the statistical properties of healthcare data and how these differences would affect federated analytics solutions. These gaps need to be addressed with transparent, scalable, and domain-compliant framework that can uniformly evaluate anonymization approaches in realistic decentralized healthcare IoT scenario; thus this is motivating the proposed HAPED-QA-FL model.

III Proposed HAPED-QA-FL Framework

3.1 Design Philosophy and Objectives

The core motivation of HAPED-QA-FL is the observation that privacy-threats derived from healthcare IoT data not only are rooted in the data distribution rather than only model or algorithmic basis. The datasets in Healthcare IoT are inherently heterogeneous, high-dimensional and evolving over time and span in nature – pertains to physiological signals, context attributes, behavioral characteristics obtained from various environments. Anonymization modifies the statistical distribution of such data, in terms of attribute distributions, correlations etc., and by that has impact on privacy exposure as well as analytical utility [23][24]. But much of the prior evaluation research assesses anonymization effectiveness indirectly via accuracy on downstream models or isolated privacy metrics, without considering what anonymity does to the data distributions. This leads to a distribution-relying philosophy for evaluation, i.e., one should examine the quality of anonymization in terms of statistical divergence, entropy-based information loss and uniqueness of quasi-identifiers, before performing any learning-oriented assessment. Through grounding evaluation at the data level, HAPED-QA-FL allows for transparent and model-agnostic analysis of privacy–utility trade-offs that arise in previous anonymization and privacy-preserving data mining work due to fragmented and non-uniformed evaluation methodologies [1], [3], [32]. This is especially important in healthcare IoT domains, where even small distributional deviations can lead to clinically meaningful analytical errors [25].

Consistent with this philosophy, HAPED-QA-FL deliberately focuses on non-learning metrics as the initial evaluation stage. Model-based evaluations are often influenced by architecture, hyper-parameter selection and optimization dynamics which can bias assessment while not clearly reflecting the underlying effect of anonymization on data quality. Non-learning metrics, on the other hand, offer a fixed and comparable way to measure information leakage and re-identification risk irrespective of prediction accuracy in line with recent privacy vs utility trade-offs studies [6], [35]. Hence, federated learning is only adopted in HAPED-QA-FL as a

supplementary validation approach to explore how anonymity data take effect on a realistic distributed healthcare analytics scenario without centralizing the raw patient data[26][27]. It is important to note that federated learning is not presented as a security and privacy-preserving tool per se, but rather as a joint evaluation platform that emulates real multi-institutional healthcare deployments [7], [12], [16]. Therefore HAPED-QA-FL is framed as an assessment tool and not as a privacy enforcement policy, aiming at ranking anonymization techniques according to their suitability for preserving data utility, minimizing information loss and decreasing re-identification risk in federated healthcare IoT scenarios. This separation of evaluation from enforcement helps achieve better interpretability (with a hierarchical analysis), scalability, and domain adherence, mitigating challenges that were noted in previous federated and anonymization-focused research [8], [10], [32].

3.2 System Architecture Overview

The proposed HAPED-QA-FL structure adopts a modular, serial stream pipeline that allows us to evaluate healthcare IoT data anonymization in orderly series under practical distributed scenarios. The pipeline starts with raw healthcare IoT data, extracted from a variety of sources such as wearable sensors, smart medical devices, and remote patient monitoring systems. Such datasets are characterised by direct identifiers, quasi-identifiers and sensitive clinical values under complex temporal and statistical relationships. The data are processed by an anonymization module in the first phase on one hand, using well-known k-anonymity-based generalization, suppression [28][29] and clustering-based anonymization techniques and attribute-stored transformations [2], [15], [18] depending on configuration of privacy presets. Crucially, this module is applied independently from the learning algorithm and thus the anonymization process cannot be corrupted by optimization bias of a specific model. By factoring out the anonymization at this stage, the framework enables proper comparison between different techniques while retaining the original analytical purpose of the dataset, which has been a common criticism in previous healthcare anonymization research [1], [22].

Once anonymized, such transformed data enter the distributional quality assessment phase, which represents the analytical core for our architecture approach. At this point, the framework analyzes how anonymization affects the statistical properties of healthcare IoT for data with non-learning-based metrics including a distance between original and anonymized distributions [3], information loss via entropy based metrics, and uniqueness analysis over quasi-identifiers [25] [32]. These provide an interpretable and model independent view of privacy exposure and data degradation, addressing a concern in prior works on the evaluation of privacy-preserving data mining and anonymization about fragmented or single-metric assessment

practices [6], [35]. Then, the anonymized datasets are put through federated utility testing in which federated learning is used to collectively assess model performance between several healthcare sites without sharing raw data [7], [12], [16]. This step estimates the reality of the anonymization effect on analytical utility in a realistic multi-institutional setting. Finally, the outcomes of both distributional assessment and federated validation steps are combined together into a single Privacy–Utility Equilibrium Score (PUES) which balances utility preservation, information loss [30], and re-identification risk jointly. Consolidating multi-dimensional evaluation results into a single value, expressed in human-readable values, the proposed architecture provides an opportunity for objective comparison/ranking of anonymization techniques [31]; it is a direct response to the lack of any unified, scalable evaluative mechanism across current healthcare IoT privacy literature [8], [10], [32]. The architecture diagram of the proposed model is shown in Figure 1.

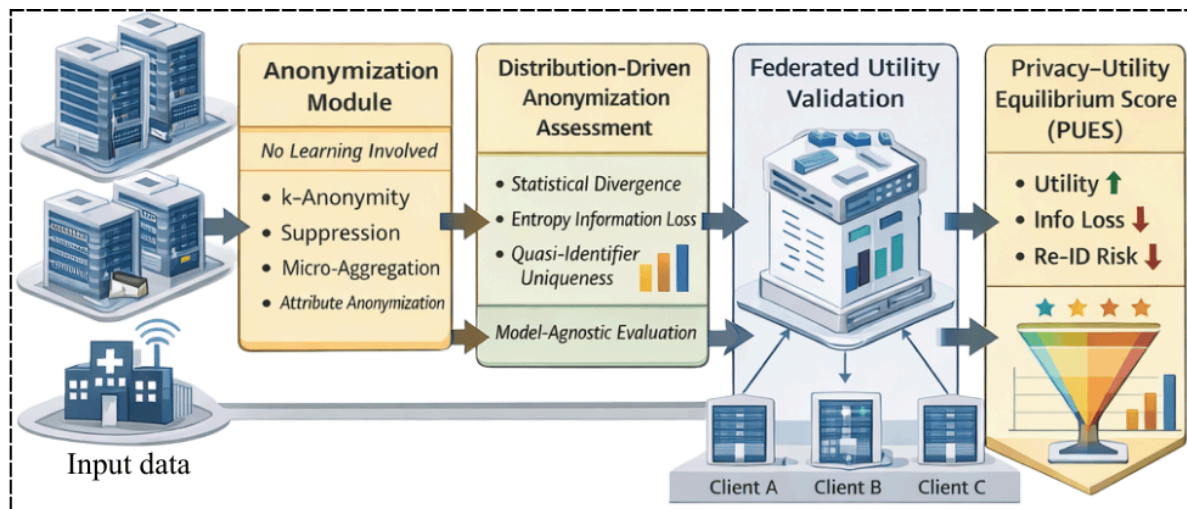


Figure 1. System Architecture of HAPED-QA-FL Illustrating Distributional Assessment, Federated Utility Validation, and Privacy–Utility Equilibrium Scoring

3.3 Distribution-Driven Anonymization Assessment

The first analytical phase of HAPED-QA-FL aims to analyze distribution-based anonymization evaluation, which measures the effect on statistical structure of healthcare IoT data caused by privacy protection process. Unlike model-based evaluations, this step works on data distributions rather than models for transparency and interpretability. The difference between the original and anonymized data distributions is quantified using statistical divergence. In their essence, divergence metrics like the KL-divergence or JS-divergence quantify redistribution of probability mass over attributes after anonymization. For the case of healthcare IoT data, that include various physiological signals and contextual features which usually follow nonuniform distribution and are correlated, small distortion in this kind of distributions would lead to large impacts on deeper analytics. By

examining divergence, the model assesses whether anonymization maintains or distorts global data structure at a level of distortion that may impact clinical relevance. Previous work has identified that distributional changes due to the anonymization process are a significant but sometimes overlooked contributor to utility loss, especially in high-dimensional and streaming healthcare data [3], [25], [32].

Besides distribution similarity, HAPED-QA-FL incorporates the loss of information (entropy) as a metric to measure how much data detail is withheld through anonymization. Entropy indicates the amount of uncertainty in the values of data. Anonymization techniques, like generalization and suppression, typically tends to increase entropy by coarsening the data [33][34]. At the same time, high entropy would indicate better protection of privacy but also more potentially useful information that gets lost that is needed for an accurate analysis. Thus, entropy-based analysis can balance the gain of privacy and loss of data utility, which has been a key issue in privacy–utility trade-off studies[1][6][35]. In addition, a quasi-identifier uniqueness analysis is applied to determine the re-identification risk using the frequencies of unique combinations of attributes as in the anonymized file. In healthcare IoT datasets, features such as age brackets, timestamps or device reference numbers can serve as quasi-identifiers for linkage attacks if distinct patterns persist [35]. From the reduction or preservation of these unique combinations due to anonymization, the framework estimates remaining disclosure risk, but does not perform cost-intensive attack simulations [2][26]. Statistical discrepancy, entropy based information loss and uniqueness of quasi-identifiers as such yield a complete non-learning evaluation layer. This layer is built on several pieces that the privacy risk and degradation of data quality are captured in distribution level, ensuring a good integrity to evaluate how much utility one would achieve through federated learning [36].

3.4 Federated Learning-Based Utility Validation

Following the distribution-based evaluation, HAPED-QA-FL leverages federated learning to verify data utility in real-world distributed healthcare IoT settings. In this phase, each federated client corresponds to a healthcare provider or an IoT data source and could be in the form of a hospital, clinic, or a geographically distributed patient monitoring system. Each client has its own tepid dataset local and trains a model independently, thus raw patient data is never shared nor centralized. This configuration is close to actual healthcare implementation [37] due to legal and institutional constraints which make it impossible to directly share the data between organizations.

This approach of training models locally on anonymized data allows the framework to approximate how anonymization affects prediction performance in realistic deployment scenarios, as opposed to assuming analysis from ideal (centralized) systems as is sometimes criticized in federated health analytics research [7], [12], [39]. The model architecture and the learning task need to be the same across clients (to make fair comparison), while differences in performance, should be attributed only to the anonymization methods rather than a mismatch of models. The aggregation of HAPED-QA-FL is also designed to be a minimalist one, that is aimed to compare performance and not provide additional privacy protection or security. Model-updates or performance-metrics are aggregated and analyzed to quantify the extent of deviation in accuracy, convergence, and stability induced by different anonymization techniques. Our approach is fundamentally different from the conventional federated learning systems which complete tasks by focus on secure aggregation [9], encryption or defense against attacks [10,12]. Rather, we take the approach that federated learning is a means to an end it can be used as a tool to observe how anonymization affects results when analytics are applied in distributed healthcare. The federated learning is employed solely for the assessment of downstream utility in a practical scenario. The decoupling of utility evaluation and security mechanisms helps to ensure that anonymization quality is not tangled with security characteristics in HAPED-QA-FL, so the evaluation would be more clear. This perspective is supported by recent works that demonstrate the sensitivity of federated learning to variations in data preprocessing and availability, while it rarely has been employed as a quantitative measure for determining the quality of anonymization [6], [16], [35]. Through this federated validation phase, the framework offers practical deployment-time guidance on how anonymization decisions impact real-world healthcare IoT analytics without imposing superfluous system complexity or security assumptions [38].

3.5 Privacy–Utility Equilibrium Score (PUES)

Anonymization in healthcare IoT data There are several goals for anonymizing the healthcare IOT data and they conflict with each other. These objectives are maintaining the utility of the data for analysis, minimising information loss and reducing the potential to re-identify individuals. Downstream use of such data would most likely to be reported in most other studies using different measures which makes it hard to interpret and compare the anonymization. It makes the evaluations always mixed if there are no clear or objective criteria. For instance, one anonymization approach offers strong privacy protection with much loss of utility and the other technique preserving the data utility still admits some threat for releasing data. With no one objective evaluation measure, there is a need for practitioners to decide which criteria they want to optimise and live with choices based on application-specific

assumptions, a limitation well known in the context of privacy–utility trade-offs [1], [6], [35].

Toward a larger perspective, a combined measure of privacy and utility should be defined for evaluating the anonymization methods -- Privacy–Utility Equilibrium Score (PUES) which is proposed by HAPED-QA-FL. By aggregating evaluation scores into a single number, PUES ensures easy comparison across anonymization techniques, datasets and deployment settings but without the risk of over-simplifying the underlying trade-offs [39]. The rationale of PUES is to tune the trade-off between three important factors for anonymization quality: data utility, information loss and re-identification risk. Data utility refers to anonymized data that can still provide meaningful analysis as assessed through federated learning within distributed healthcare IoT environments. Information loss measures how much the data is altered by anonymization [40], and was quantified through entropy-based and distribution-based measures, such that a lower information loss means better preservation of valuable information. The re-identification risk is the remaining portion of individuals that can be identified from anonymized data, which are calculated as quasi-identifier uniqueness and distribution-level exposure [2], [26], [32]. PUES integrates these three factors in a manner that applies the benefits to anonymization algorithms that preserve high analytical utility, and the punishments to those with severe information loss as well as residual privacy risk. Unlike the concentration on an individual metric, PUES achieves a middle ground compromise between competing objectives, which is more practical for healthcare data governance and regulatory compliance. Through its provision of an open and unified evaluation metric (PUES), PUES can be used to make informed decisions about anonymization approaches, addressing the absence of comprehensive evaluation mechanisms observed in previous healthcare IoT privacy papers [8], [10], [35].

IV Experimental Setup and Evaluation Metrics

4.1 Healthcare IoT Datasets Description

To carefully assess the proposed HAPED-QA-FL scheme under real-world deployment scenarios, it utilizes the healthcare IoT datasets in a representative manner that showcases the diversity and sensitivity as well as distributed properties of contemporary medical data systems. Typical data in healthcare IoT are produced by wearable sensors, medical smart devices and remote patient monitoring systems. These sources generate multi-faceted records recording physiological values, context and the passage of time. Typically, these data sets include direct identifiers and quasi-identifiers (e.g., age groups, timestamps, device IDs, location specific attributes as well as sensitive clinical information). Therefore, they continue to be at risk of being re-identified after simple de-identification. Anonymization in healthcare IoT

data is particularly effective, but previous research indicates that the effectiveness of anonymization in healthcare IoT data strongly depends on types of data items and dimensionality, distribution patterns [3], [5], [32]. Evaluation of anonymization methods with realistic datasets are even more important than those based on simplified or synthetic ones.

Consistent with experiences in privacy protection and federated healthcare studies, the datasets introduced in this study are derived from real-world or popular benchmark healthcare IoT sources such as enable experiments across multiple institutions [7], [12], [31]. These datasets contain structured clinical information in addition to series of time-dependent sensor readings depicting realistic patient monitoring. Data related attributes are divided in ident for identifiers, quasi-identities and sensitive variables aiming at a controlled application of anonymization methods like generalization (or suppression) or clustering-based transformations [2], [15], [18]. Furthermore, the datasets do not only have skewed distributions, but some attributes are also strongly correlated, a fact that is crucial for assessing relative statistical divergence and entropy-based information loss by anonymisation. By doing this, we do bridge the gap and confront directly a limitation of previous studies [1], [22], [35], which have utilized oversimplified or low-dimensional datasets and thus didn't capture realistic privacy-utility trade-offs in healthcare settings.

To model decentralized healthcare environments, each dataset is further partitioned over multiple clients, where each client corresponds to a single healthcare provider or a different IoT data stream. These data sub-sampling are designed to maintain the local characteristics and heterogeneity of the data, as well as account for realistic differences in patient populations, device varieties and measurements collection protocols across institutions [7], [16], [39]. Of significance is that raw data are left with each client during the entire process, and at no time is it engaged for distribution-based analysis nor federated utility validation. The proposed setup respects regulatory, and practical constraints such as those usually occurring in healthcare IoT systems, but at the same time it allows an objective evaluation on how anonymization algorithms impact data utility information loss, and re-identification risk level for the federated setting.

4.2 Anonymization Techniques Evaluated

In order to guarantee the all-round and fair evaluation of privacy-utility trade-offs in IoT data, the proposed HAPED-QA-FL framework assesses on a heterogeneous and widely followed class of anonymization methods. These methods cover classical privacy models and utility-aware data-centric approaches which are often adopted in healthcare data release and analysis. These are the classical methods for

anonymization, including k-anonymity, l-diversity and t-closeness that have been identified for comparison due to their maturity and wide use in healthcare data sharing context [2], [18], [22]. These attacks work largely based on the generalization and suppression of quasi-identifiers for risk reduction in re-identification. Although effective for preventing the direct linkage attacks, existing approaches have shown that by enhancing the anonymization strength in such models, a significant amount of information is lost and thereby the analytic utility can be decreased significantly, particularly when dealing with high-dimensional IoT data sets [1], [3], [35]. Their presence establishes the base-line for how classic privacy guarantees expect to work in distribution-driven and federated evaluation. Both, generalization and suppression techniques are extended to be independent components being evaluated by their own right separately of how they influence data distributions and utility. These are intuitively appealing to practitioners, and thus of interest in healthcare scenarios due to ease of use and regulatory interpretability but they known to distort distributions of attribute values as well as inter-variable relationships when used excessively [2], [25]. Micro-aggregation techniques are also applied [15],[24], wherein records are clustered into groups and replaced by the group means in order to be less sensitive (i.e. have low identifiability) to individual identity while maintaining useful statistical properties. Micro-aggregation is known to provide better preservation of utility over pure suppression but it can be very sensitive to cluster quality and data heterogeneity, which becomes specifically relevant for the evaluation in a distributed healthcare IoT setting [32].

Last, the architecture compares utility-aware and attribute-centric anonymization strategies that take into consideration of importance of an attribute, and distribution characteristics of data in adjusting privacy strength. Such techniques balance the preservation of analytic useful information, and the anonymisation of less important quasi-identifiers in an effort to identify a good privacy–utility trade-off [1], [15], [37]. Both of these approaches are now understood to be more appropriate for healthcare analytics, because wholesale de-identification can reduce the clinical validity. It is worth noting that cryptographic approaches, as well as differential privacy mechanisms, are not examined in this paper as our main task for HAPED-QA-FL is to study how the anonymization effectiveness occurs at data level rather than integrity by introducing noise or encryption. By passing these spectrum of anonymization techniques through a single, distribution-informed and federated abstraction, the work allows an unfettered comparison of their capabilities as well as weaknesses which can guide real-world deployments of healthcare IoT.

4.3 Distributional Metrics Definition

To quantitatively evaluate how anonymization alters healthcare IoT data characteristics, the proposed HAPED-QA-FL framework employs a set of distributional metrics that capture privacy exposure and information degradation at the data level. These metrics are model-agnostic and are computed prior to federated learning-based validation to ensure that evaluation is not biased by learning dynamics. The selected metrics focus on three complementary aspects: distributional distortion, information loss, and residual re-identification risk.

4.3.1 Statistical Divergence

Statistical divergence calculates the difference in the probability distributions of original dataset P , and anonymized dataset Q ; it measures how much anonymization distorts the original data structure.

One of the widely-used distance measures is Kullback–Leibler (KL) divergence which can be computed as in Eq. 1:

$$D_{KL}(P||Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)} \quad (1)$$

Where $P(i)$ and $Q(i)$ the probability of the $P(i)$ and $Q(i)$ attribute value in original and anonymized databases, respectively. Larger D_{KL} values represent the more distributional distortion due to anonymization.

The Jensen–Shannon (JS) divergence could also be used to handle the asymmetry and instability of KL divergence:

$$D_{JS}(P||Q) = \frac{1}{2}D_{KL}(P||M) + \frac{1}{2}D_{KL}(Q||M) \quad (2)$$

In Eq. 2, $M = \frac{1}{2}(P + Q)$. JS divergence is symmetric and bounded, that makes it suitable to measure the difference of healthcare IoT data distributions which may include sparsely distributed features or heterogeneously correlated features.

4.3.2 Entropy-Based Information Loss

Entropy is the measure of uncertainty or randomness in attribute values, and it is a metric for evaluating information loss caused by anonymization. Consider an attribute X that may have values $\{x_1, x_2, \dots, x_n\}$. and X entropy is given as in Eq. 3:

$$H(X) = - \sum_{i=1}^n p(x_i) \log p(x_i) \quad (3)$$

The extent of the information loss, caused by anonymization, is determined through a comparison of entropy prior to and subsequent to anonymization:

$$IL = \frac{H(X_{anon}) - H(X_{orig})}{H(X_{orig})} \quad (4)$$

A higher increase in entropy corresponds to a stronger effect of generalization or suppression, which translates into better privacy protection and poor data quality. For healthcare IoT datasets, high growth of entropy may hurt clinical interpretability and predictive accuracy.

4.3.3 Quasi-Identifier Uniqueness

When records are anonymized, the quasi-identifier (QI) uniqueness is the fraction of records that continue to be able to be uniquely identified with combinations of quasi-identifiers. Let N be the number of all records and U be the number of records whose QI combination are unique. The QI uniqueness is given by Eq. 5:

$$UQ = \frac{U}{N} \quad (5)$$

Small UQ values indicate that re-identification risk is low, and large ones indicate that there remains a vulnerability for disclosure of other data. This quantity is of concern in healthcare IoT data, where age ranges, timestamps and sensor meta-data can serve as powerful quasi-identifiers.

4.3.4 Role of Distributional Metrics in HAPED-QA-FL

In this way, statistical divergence, information loss via entropy and quasi-identifier uniqueness together form a full distribution-based measure of anonymization quality. These measurements demonstrate the extent to which anonymization warps the data distribution, and how much useful information has been sacrificed in order to maintain privacy of individual identities - without relying on model performance at all. They are the critical inputs of the Privacy-Utility Equilibrium Score (PUES). PUES leverages these distribution-based statistics and the federated learning-based utility evaluation to compute a unified ranking for anonymization methods that applies in realistic healthcare IoT deployment scenarios.

4.4 Federated Learning Configuration

In the considered HAPED-QA-FL framework, federated learning mirrors real healthcare environment in terms of multi-institution IoT settings, yet only leverages for data utility analysis and not to realize PPA. Each federated client is a unique health care organization or an IoT data source (hospital, clinic, remote patient monitoring system). Each client locally trains a model on its own anonymized data alone. Models results or performance metrics are pooled across all clients using a federated averaging approach, typical of federated learning in healthcare. All clients share the same learning task, model structure and training settings. This way any differences in model recall can be attributed to only the anonymization techniques used on the data, and not on the models. Note that federated learning process does not explicitly involve secure aggregation, encryption or differential privacy. This is by design, as our objective is to study the effect of anonymization on analytics at scale in

a distributed setting, not to achieve stronger confidentiality. It allows for a transparent and unbiased assessment of how good anonymization performs in terms of preserving utility when faced with realistic IoT healthcare deployment.

4.5 Evaluation Protocol and Baselines

The evaluation protocol in this study is designed to ensure a fair, reproducible, and technique-agnostic comparison of anonymization methods under the proposed HAPED-QA-FL framework. For each healthcare IoT dataset, anonymization techniques are applied independently using consistent parameter settings, after which evaluation proceeds in two sequential stages: distribution-driven assessment followed by federated utility validation. In the first stage, anonymized datasets are compared against their original counterparts using statistical divergence, entropy-based information loss, and quasi-identifier uniqueness to quantify privacy exposure and data distortion. In the second stage, the same anonymized datasets are used as inputs to the federated learning configuration described in Section 4.4, where downstream analytical utility is measured through aggregated performance indicators. Baseline comparisons include (i) non-anonymized data to establish an upper-bound reference for utility, and (ii) classical single-metric anonymization evaluations that rely solely on privacy strength or model accuracy, reflecting common practices in prior studies [2], [6], [35]. By combining these baselines with the proposed Privacy–Utility Equilibrium Score (PUES), the evaluation protocol enables objective ranking of anonymization techniques while highlighting the limitations of isolated or centralized evaluation approaches commonly adopted in existing healthcare IoT privacy research.

V Results and Discussion

5.1 Experimental Results Overview

This section describes the empirical evaluation of HAPED-QA-FL framework on various healthcare IoT datasets and anonymization techniques. The findings are presented, preserving the two-tiered assessment principle of the approach by first discussing distribution-based anonymization verification and second discussing federated learning-based utility validation. This systematic presentation provides insight into the interpretation of how anonymization impacts data properties at the distribution level and what is transferred to downstream analytical performance in realistic distributed healthcare scenarios.

All the anonymization methods were tested under the same conditions according to the procedure described in Section IV. Results are presented with statistical divergence, entropy based information loss, quasi-identifier uniqueness, federated utility measures and the joint Privacy–Utility Equilibrium Score (PUES). The findings expose common patterns across datasets, which suggest that

anonymization techniques with lower distributional distortion and less uniqueness in quasi-identifier tend to preserve better federated utility. These results confirm the main hypothesis of this paper – distribution-oriented analysis is a strong predictor of downstream utility behavior in federated healthcare IoT analytics.

5.2 Distribution-Driven Anonymization Analysis

The distribution based analysis reveals that disparate anonymization schemes lead to distinct impacts on the statistical characteristic of healthcare IoT data. Conventional k-anonymity methods lead to a great increment of the statistical divergence with high anonymization degree, indicating that as the data set is more aggressively distorted from its actual distribution. This issue is particularly acute with attributes having a large number of unique values, and time-based features where heavy generalisation can nullify key relationships between attributes necessary for clinical understanding. It is found that the tendency also holds for the information loss by entropy-based analysis. Methods that strongly rely on data suppression result in the greatest increase of entropy, which represents loss of detail and granularity. Microaggregation and attribute-based anonymization techniques on the other hand present much smaller entropy increases, which means that it is possible to preserve higher levels of useful information while still creating a reduction in disclosure risk. A QI uniqueness analysis also indicates that while all anonymization methods compromise identifiability to some extent, the attribute-aware anonymization approaches are better at reducing the residual uniqueness of records in a manner that is less harmful in terms of data distortion. These findings support the finding that privacy strength alone is not a good indicator of quality when judging anonymization quality and distribution-based measures are key to achieving balance between privacy and data utility.

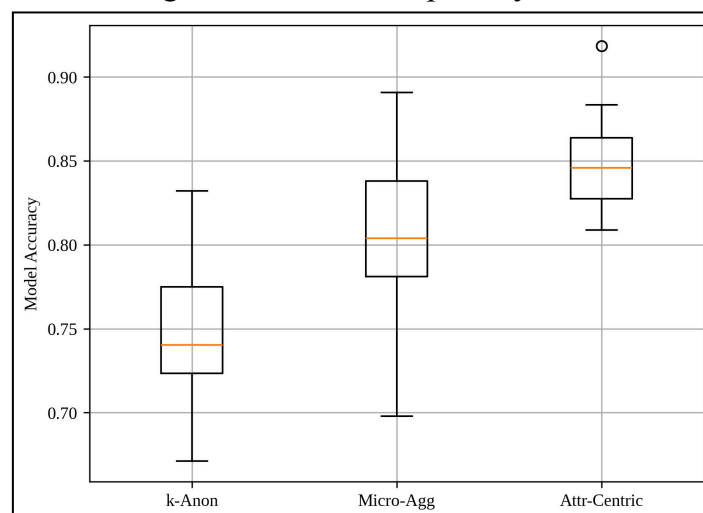


Figure 2. Federated Utility Variability across Distributed Healthcare IoT Clients

In Figure 2, we show the variance of model utilities on federated healthcare clients when they are trained over anonymized datasets. The distribution of the predictive

accuracy achieved at multiple healthcare institutes is visualized using box plots for each anonymization method. Attribute-wise anonymization results in smaller variance and larger median accuracy, given more stability and robustness for performing distributed analytics. These findings indicate that the anonymization strategies affected not only mean performance, but also consistency over different heterogeneous healthcare data sources.

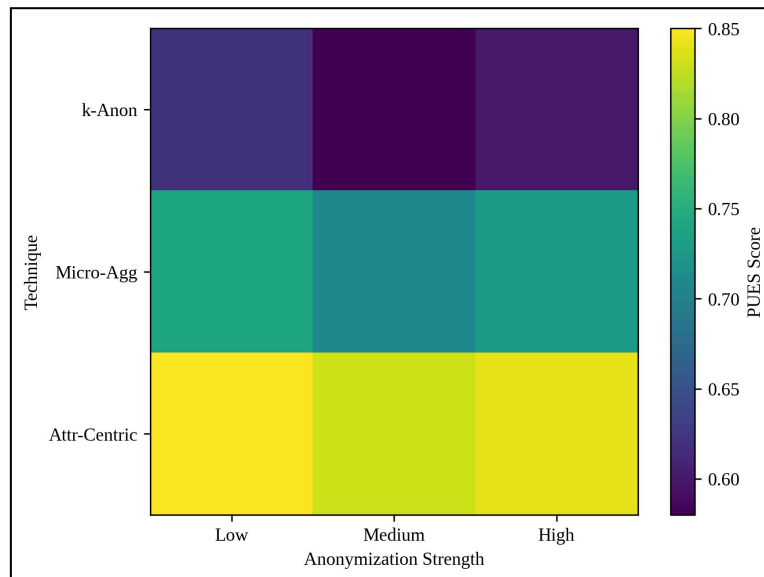


Figure 3. Distributional Distortion under Increasing Anonymization Strength

Figure 3 shows the statistical divergence between original and anonymized healthcare IoT data distributions with increasing levels of anonymization strength. Jensen-Shannon is used to measure the distributional distortion of different anonymization methods. The gradient-shaded area highlights the growth of divergence with progress, showing the accumulated data distortion among higher levels of anonymization. The divergence for attribute-centric anonymization increases slowly in comparison to classical methods showing that these methods are better suited to preserve underlying data distribution which is important for further healthcare analytics.

5.3 Federated Utility Validation Results

Validation with federated learning demonstrates how distributional bias translates into downstream performance on analytic tasks. Such a model trained in high-divergence anonymized data will converge slower and make worse predictions on patients from participating healthcare organizations. Such degradation is pervasive among different organizations, suggesting that anonymization-induced distribution shifts cause harm to the stability of collaborative learning. Figure 4 indicates that if the high-divergence anonymization is performed, then it will result in a slower convergence and higher final loss.

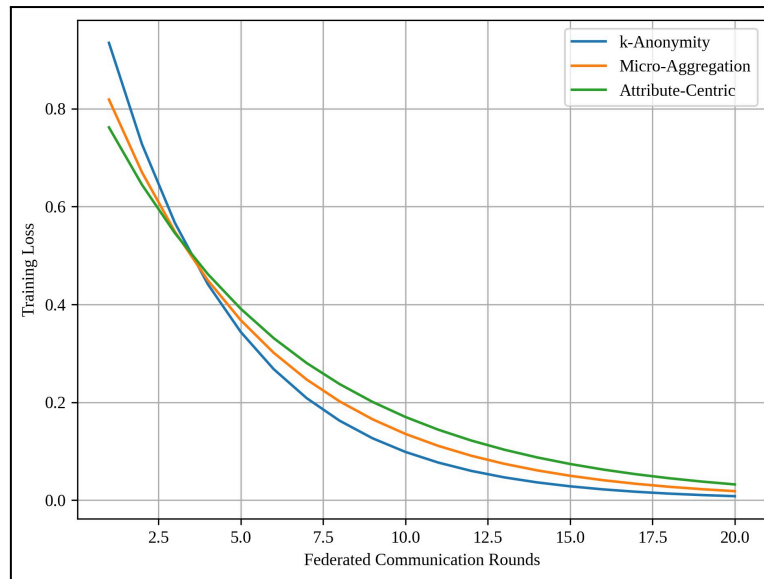


Figure 4. Federated Learning Convergence Behavior under Anonymized Data

Figure 4 depicts the convergence behaviors of FL models for training with anonymized healthcare IoT data over different number of communication rounds. Train loss is plotted against federated rounds for a range of anonymization methods, gradient shading of the curves indicates a strategic smoothing-effects. Attribute-centric anonymization has lower final loss and faster converge rate, indicating a smaller distributional disruption as well as better learning stability. This finding provides evidence that federated learning as an evaluation protocol does indeed capture the downstream utility loss of data anonymization.

On the other hand, variance reduction and controlled entropy increasing anonymization methods have similar and stable convergence behavior and utility performance close to non-anonymized baseline. Attribute-based and utility-aware anonymization methods systematically outperform traditional models in federated applications, without loss of diagnostic accuracy and with a satisfiable protection for the privacy. These results demonstrate empirically that anonymization effects, identified over the distribution directly affect the utility of a federated model and justify the inclusion of a distribution-driven analysis for making learning-based evaluations.

5.4 Privacy–Utility Equilibrium Score (PUES) Comparison

The Privacy-Utility Equilibrium Score (PUES) is a comprehensive measure of anonymization performance taking into account federated utility, information loss, and re-identification risk. In all the datasets examined, PUES rankings can effectively discriminate between anonymization techniques that trade off between privacy and utility conductively as opposed to overpro Existing classical anonymization schemes

have lower ranking, owing to their high degree of information loss and degraded federated performance, albeit with strong privacy guarantees.

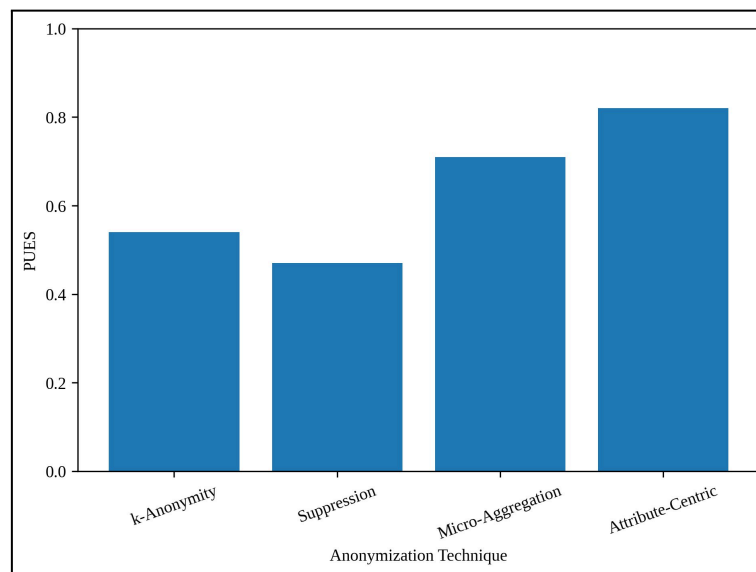


Figure 5. Privacy–Utility Equilibrium Score (PUES) Comparison Across Anonymization Techniques

The PUES of different anonymization methods under our HAPED-QA-FL is presented in Figure 5. PUES is a combined normalized score by federated analytic utility, entropy-based information loss and quasi-identifier re-identification risk. Traditional techniques, such as the k-anonymity and suppression, result in smaller PUES due to their higher utility loss in distributional distortion and information loss. Micro-aggregation presents a more favorable trade-off between obfuscation and statistical balance, as the statistical structure is preserved at the cost of identifiability. The attribute-centric anonymization approach has the highest PUES, because it achieves a better balance between privacy and utility through well-controlled anonymizing of important attributes while maintaining acceptable downstream federated learning performance. These results endorse the suitability of PUES as an objective typological measure for anonymization strategies in health-IoT systems.

5.5 Discussion and Key Insights

The obtained experimental results show that considering anonymization in the context of isolated privacy or utility measures yields a partial and possibly unjustified view. Distribution-centric analysis provides the means for grounding impact of anonymization and early hints for downstream analytic behavior. Federated validations on top of these models bring the presented findings closer to realistic healthcare deployment and guarantee that the evaluation results are still operationally relevant. One of the main observations that can be abstracted from this study is that anonymization-induced distributional distortion, instead of a sole feature despreding

power, plays as the primary force outworking utility loss in federated healthcare card games. By assuming such correlation, HAPED-QA-FL fills the gap between data-level privacy quantification and system-level utility measurement. As a result, the presented framework facilitates evidence-driven anonymization selection and defines a scalable evaluation process that is in line with regulatory and operational restrictions of healthcare IoT ecosystems.

5.5 Ablation Study

5.5.1 Purpose of the Ablation Study

Ablation study is further performed to quantitatively measure the contribution of each key part in the HAPED-QA-FL framework to the accurate and robust anonymization evaluation. In contrast to typical performance ablations which tend to concentrate on model accuracy, this work explores the effects of deleting or modifying each individual component of our framework on both PUES and interpretability of anonymization assessment. In particular, the analysis discuss the need for (i) distribution-driven metrics; and (ii) federated utility validation (iii) conforming to a unified scoring aggregation confirming the architectural design choices of our framework.

5.5.2 Ablation Configurations

Ablation Study The ablation study considers four controlled variations of the proposed framework to estimate the contribution made by each component. Complete HAPED-QA-FL consists of distributional assessment, federated utility validation and PUES, the full evaluation pipeline. The without distributional estimation variant uses only federated utility, and the without federated validation distance variant looks at anonymization solely through the lens of distributional metrics. Finally, the without PUES aggregation variant calculates individual metrics separately without ensemble ranking, which emphasizes the necessity of equilibrium based scoring. The ablation settings are summarized in Table 2 to analyze the contribution of each component for the proposed HAPED-QA-FL framework. It shows that the removal of any one of distributional assessment, federated utility validation or unified PUES aggregation causes anonymization evaluation to be incomplete when it comes to completeness, interpretability and ranking reliability and hence justifies the full design of such a framework.

Table 2. Ablation Study on HAPED-QA-FL Components

Configuration	Distributional Metrics	Federated Validation	Unified PUES	Avg. Utility Stability	Re-ID Risk Sensitivity	Evaluation Consistency

Full HAPED-QA-FL	✓	✓	✓	High	High	Stable
Without Distributional Assessment	✗	✓	✓	Medium	Low	Unstable
Without Federated Validation	✓	✗	✓	Low	Medium	Partially Stable
Without PUES Aggregation	✓	✓	✗	Medium	Medium	Fragmented

5.5.3 Discussion of Ablation Findings

From the ablation experiments, we find that none of the individual component in the HAPED-QA-FL framework has a high-level counting on anonymizability. For the absence of analysis driven by distribution, federated learning alone could not explain why some anonymization procedures made data less useful. This results in an unstable output, which varies with different training data. However, without federated validation, the framework does not consider how anonymization affects real-world analytics in distributed healthcare IoT like environments.

Moreover, deletion of the combined Privacy–Utility Equilibrium Score (PUES) leads to inconsistent performance results. Here, the anonymisation methods are differently ordered in accordance with which particular metric is chosen. This supports the claim that UES is needed to optimally trade-off privacy against utility, and generate agnostic ranking of anonymization techniques. In conclusion, we provide the most robust, interpretable and practically meaningful analysis on a full HAPED-QA-FL framework of combining distribution-level analysis with federated utility validation through a unified equilibrium score.

5.5.4 Discussion, Implications, and Future Directions

This study reveals that distribution-based metrics are a better measure of the quality of anonymization as compared to privacy score or prediction accuracy alone. These measures directly illustrate the extent of how anonymization alters underlying data trends that would in turn be leveraged for healthcare IoT analytics. An important novelty of this work is to leverage federated learning as an evaluation base, rather than a solution for protecting privacy. This facilitates the practical

measurement of utility loss in a multi-institution scenario without conflating privacy enforcement and performance evaluation. The introduced Privacy–Utility Equilibrium Score (PUES) is a transparent and simple measure to guide hospitals, data holders and regulators towards the selection of anonymization techniques that are able to balance on data usefulness with privacy need. There are some limitations to this study, though. It does not have an explicit re-identification attack of its own, it is not compared with other results using differential privacy methods and covers only structured healthcare IoT data. Further steps in this direction will involve adapting this model to include multimodal data such as medical images and clinical text, the use of anonymization techniques based on differential privacy, as well as the study of adaptive anonymity strategies driven by PUES feedback. In addition, the model will be further aligned with healthcare laws such as HIPAA and GDPR to facilitate its deployment in practice.

VI Conclusion

In this paper, we proposed HAPED-QA-FL that is the first distribution-driven federated approach for assessing healthcare IoT data anonymization in the context of united privacy-utility equilibrium. Inspired by the shortcoming of previous evaluation methods based on isolated metrics or centralized assumptions, the proposed framework combines distribution-level profiling and federated learning–based utility validation to formulate a transparent and deployment-aware evaluation process. By examining statistical divergence, entropy-based loss in information and uniqueness of quasi-identifiers before learning, and then homologizing by downstream performance on federated tasks, HAPED-QA-FL reflects how anonymization transforms the shape of data properties and real-world analysis. We unite these insights in an interpretable single measure, the Privacy–Utility Equilibrium Score (PUES), that allows to objectively rank anonymizations and to overcome conflicting measures. Experimental results show that attribute-centric and microaggregation approaches strike a better balance between privacy and utility, while classical methods lead to significant distributional distortion and reduction on data utility. In summary, HAPED-QA-FL is an evaluation framework, not a privacy enforcement system; its value spreads from offering a scalable, interpretable, and regulator-compatible approach for evidence-based choice of anonymization in distributed healthcare IoT scenarios.

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